

Athresh Karanam

 athreshkaranam

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 athresh

 Work Authorization: F1

Education

University of Texas at Dallas

M.S. and Ph.D., Student in Computer Science, Advisor: **Prof. Sriraam Natarajan**

Aug. 2018 – May. 2026

Dallas, TX

Indian Institute of Technology, Guwahati

Bachelor of Technology in Civil Engineering, Advisor: **Prof. Subashisa Dutta**

July 2011 – May 2015

Guwahati, India

Research Interests

Deep Tractable Probabilistic Models, Knowledge-based systems, Diffusion Models, Data and Compute Efficient Learning, Discrete Optimization, Generalizable ML, Causal Inference, and AI for Healthcare.

Publications

Karanam, A., Mathur, S., Sidheekh, S., Natarajan, S., A Unified Framework for Human-Allied Learning of Probabilistic Circuits: In **AAAI 2025**

Mathur, S., Sidheekh, S., **Karanam, A.**, Natarajan, S., Deep Tractable Probabilistic Models: Tutorial at **CODS-COMAD 2024**

Karanam, A., Natarajan, S., Test-time Active Feature Selection through Tractable Acquisition Functions: In **TPM Workshop at UAI 2023**

Karanam, A., Mathur, S., Sidheekh, S., Natarajan, S., Bayesian Learning of Probabilistic Circuits with Domain Constraints: In **TPM Workshop at UAI 2023**

Mathur, S., **Karanam, A.**, Radivojac, P., Haas, D.M., Kersting, K., Natarajan, S., Exploiting Domain Knowledge as Causal Independencies in Modeling Gestational Diabetes: In **PSB 2023**

Karanam, A., Killamsetty, K., Kokel, H., Iyer, R., ORIENT: Submodular Mutual Information Measures for Data Subset Selection under Distribution Shift: In **NeurIPS 2022**

Karanam, A., Mathur, S., Radivojac, P., Kersting, K., Natarajan, S., Explaining Deep Tractable Probabilistic Models: The sum-product network case: In **PGM 2022**

Zečević, M., Dhimi, D. S., **Karanam, A.**, Natarajan, S., and Kersting, K., Interventional sum-product networks: Causal inference with tractable probabilistic models: In **NeurIPS 2021**

Karanam, A., Hayes, A.L., Kokel, H., Haas, D.M., Radivojac, P., Natarajan, S., A Probabilistic Approach to Extract Qualitative Knowledge for Early Prediction of Gestational Diabetes: In **AIME, 2021.**

Technical Skills

Python, PyTorch, Java, MATLAB, R, Git, SQL.

Experience

AI Intern, Altir LLC

Sept. 2025 – Nov. 2025

Delivered an interpretable, uncertainty-aware LLM agent by augmenting models with local probabilistic reasoning, improving query-answering accuracy and reliability through faithful uncertainty quantification.

Research Assistant, Starling Lab, UT Dallas, TX

Aug. 2020 – Present

Working on tractable probabilistic models, data subset selection, explainability, incorporating expert knowledge into ML models, and their applications to healthcare data.

Tutorial on TPMs, IIT Madras

Mar. 2023

Presented a 3-day tutorial on Tractable Probabilistic Models to researchers and ML practitioners at IIT Madras. Taught jointly with Prof. Sriraam Natarajan, Saurabh Mathur and Sahil Sidheekh.

Instructor

Aug. 2021 – Oct. 2021

Organized an 8-Week course of Advanced Machine Learning for engineers from Vistra Corporation. Taught jointly with Prof. Natarajan, Sriraam and Kokel, Harsha.

ML Intern, RelationalAI Inc.

Jan. 2020 – Apr. 2020

Used tractable probabilistic models to build an interactive tool for compatible product recommendation. Trained and evaluated boosted relational models on online vendor's catalogue.

Software Developer, NetCracker Technology Pvt. Ltd.

June 2015 – Jan. 2017

Worked on a product called ICOMS (Integrated Communications Operations Management system). Developed new features under several ICOMS releases.

Projects**Human Allied Learning**

May. 2024 – Present

Developing methods to elicit and integrate knowledge from domain experts into deep generative probabilistic models.

Data subset selection for compute-efficient domain adaptation

May. 2021 – Jun. 2022

Developed a model training framework for compute-efficient supervised domain adaptation (SDA) through data subset selection. Resulted in $2\times$ to $3\times$ speed-up in training times of existing SDA techniques.

Generative models for nuMoM2b medical data

Aug. 2020 – Present

Building probabilistic generative models for a cohort from a healthcare domain. Working on effective modeling, explainability, and efficient inference.

Causal Inference with tractable probabilistic models

Nov. 2020 – May. 2021

Helped formalize and implement a new technique to learn interventional sum-product networks capable of tractable causal inference. Conducted ablation study to validate robustness of the model.

SRL model for credit default risk prediction

Aug. 2019 – Nov. 2019

Trained and evaluated a statistical relational model for Kaggle Home credit default risk dataset and extracted first order rules from it to explain predictions.

Academic Services

- Volunteered and reviewed papers for AAAI 2022-25, ICLR 2024, NeurIPS 2023-2026, Neuro Causal and Symbolic AI workshop at NeurIPS 2022, Data Mining and Knowledge Discovery Journal, Journal of Artificial Intelligence Research